

Research Statement

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Research Agenda

My research lies in experimental economics, behavioral economics, and game theory. I study how information and institutional design shape communication, cooperation, and coordination. My central question is when changes in incentives and information produce the behavior they are designed to elicit, and why they sometimes fail. I combine game-theoretic models with controlled experiments to identify these obstacles and the institutional changes that can address them.

The obstacles take different forms. A punishment that disciplines an informed sender may also discourage a receiver from trusting. Potential victims may value collective resistance yet remain inactive when they face an uncertain common threat. A contract that lowers the theoretical cost of motivating teamwork may require workers to make difficult inferences about one another's incentives. In each case, a design that works in theory can fail in practice because it relies on beliefs, inferences, or responses that participants do not supply.

My job market paper studies how punishment and role uncertainty shape honesty and trust in repeated communication. Two further projects examine the informational foundations of collective resistance and the cost-effectiveness of team incentives under different forms of observability and pay transparency. These three projects share a common theme: an institution's effectiveness depends on the responses of multiple participants, so improving one person's incentives need not secure another's response. In each of these projects, theory identifies the incentives and beliefs an institution requires, and experiments test whether behavior meets those conditions. A fourth project examines how individuals evaluate outcomes against unrealized personal and social alternatives. Together, these projects form a research agenda on *Communication and Coordination under Incomplete Information: Theory and Experiment*.

Honesty and Trust in Repeated Communication

Repeated professional exchanges, such as recommendation letters and investment referrals, require both credible reports and recipients willing to act on them; the prospect of future interaction need not secure both. My job market paper, *Role Uncertainty and Punishment Severity in Repeated Cheap Talk: Theory and Experiment*, coauthored with Wooyoung Lim and under revise-and-resubmit at *Games and Economic Behavior*, studies how punishment and role uncertainty affect the sustainability of truthful communication, and whether they improve honesty, trust, and efficiency in the laboratory. I originated the research question and led the theoretical analysis, the experimental design and implementation, and the data analysis.

Model and Theoretical Predictions

We study an indefinitely repeated cheap talk game. In each round, an informed sender privately observes the state and sends a message. The receiver then chooses one of two state-contingent actions or a safe action with a state-independent payoff. The sender always prefers one particular action, whereas the receiver prefers to match the action to the state. In the one-shot game, communication is uninformative and the receiver chooses the safe action. Repetition can instead sustain truthful communication through history-dependent strategies.

Our main theoretical benchmark is a truth-telling equilibrium: the sender reports truthfully, the receiver follows the message, and any deviation triggers permanent reversion to the one-shot outcome. We vary two institutional features. Greater *punishment severity* lowers the sender's payoff following a breakdown of communication but also raises the receiver's payoff from the safe action. Under *fixed roles*, participants remain sender and receiver throughout the relationship. Under *role uncertainty*, they may occupy either position over time.

Relative to fixed roles with mild punishment, each feature can facilitate this equilibrium when introduced separately. With fixed roles, harsher punishment strengthens the sender's incentive to be honest. Under mild punishment, role uncertainty lets both participants share the informational surplus of the receiver role.

However, these two features need not reinforce each other. Under role uncertainty, punishment severity becomes a double-edged instrument: harsher punishment lowers a participant's punishment payoff as a sender but raises it as a receiver. With equally likely roles, these effects cancel in expectation. With unequal role probabilities, the binding incentive constraint can shift between participants as punishment becomes harsher, so moderate punishment can facilitate truth-telling while excessive punishment can hinder it.

Experimental Design and Findings

We test these predictions in six treatments that cross mild or harsh punishment with fixed roles, unequal-chance random roles, or equal-chance random roles. We measure sender truthfulness, receiver trust, and communication efficiency separately. This decomposition is essential: high efficiency need not imply that information is transmitted, and greater honesty need not improve efficiency if receivers remain unwilling to act on messages. Three patterns show that equilibrium sustainability alone cannot fully account for behavior.

First, with fixed roles, harsh punishment increases sender truthfulness but not receiver trust or efficiency; receivers instead choose the safe action more often. We call this the *safety trap*: the payoff change that disciplines senders also makes the receiver's fallback more attractive, raising the opportunity cost of acting on a message. An instrument designed to solve the sender's incentive problem thus creates a new obstacle on the receiver's side.

Second, under mild punishment, role uncertainty tends to increase trust without a comparable increase in honesty. This pattern is consistent with a *blind trust equilibrium*, in which senders always send the message that induces their preferred action and receivers follow it even though it conveys no information about the state. Because role rotation lets participants share the benefits of the sender role, this outcome can be efficient by our surplus measure without transmitting information.

Third, under role uncertainty, harsher punishment reduces trust despite somewhat greater sender honesty. The higher safe payoff appears to make participants conservative as receivers even when they are more honest as senders, so the same participant behaves differently across roles. Within-game evidence supports this interpretation: receivers often choose safety from the first round or abandon established cooperation without a preceding lie, so their use of the safe action is not merely punishment for past dishonesty.

The project shows that honesty, trust, and efficiency need not move together. Truthful communication is a demanding form of cooperation: it requires both disciplined senders and receivers confident enough to act on their reports. Blind trust is less demanding, because it requires no verification, but it tolerates uninformative messages. An institution may therefore encourage accurate reporting without encouraging receivers to use those reports, or sustain surplus-generating cooperation without informative messages. A system meant to elicit reliable information must be evaluated differently from one meant only to sustain mutually beneficial interaction.

The Informational Foundations of Collective Resistance

Sequential aggression often succeeds even when potential victims would benefit from collective defense. Standard explanations emphasize conflicting interests, free riding, or failures to coordinate simultaneous action. In *First They Came for the Others: A Theory of Divide-and-Conquer*, coauthored with Yeon-Koo Che and Wooyoung Lim, we identify a distinct informational friction: a bystander who observes an attack on someone else must infer whether it reflects a target-specific motive or the beginning of a broader campaign that will reach her.

In the model, the attacker privately observes the benefits of conquering two potential victims and can attack them sequentially. The first victim decides whether to resist; if she does, the second victim observes

the attack and the resistance decision before deciding whether to join. A bystander who knew she would also be targeted would join, and anticipating this support, the first victim would fight. Any failure of collective resistance therefore stems from uncertainty about the scope of the attacker's motive rather than from a simultaneous-move coordination problem.

Two canonical equilibrium outcomes emerge. In a *divide-and-conquer* equilibrium, the initial attack appears sufficiently localized that the bystander stays out; anticipating no support, the first victim surrenders, and the attacker proceeds sequentially. In a *collective-resistance* equilibrium, the same attack is read as evidence of a broader campaign: the bystander joins, the first victim fights, and only attacker types willing to face joint resistance attack at all.

Several results run against a simple cost-based intuition. A higher cost of aggression can make divide-and-conquer easier to sustain, because a costly attack may signal an unusually strong motive against the initial target rather than an intention to attack both victims. Lower correlation between the victims' fates likewise makes the first attack appear more localized. We also show how wishful and magical thinking, particularizing rhetoric, neutrality and collective-defense treaties, and alliance networks reshape this inference. Effective collective defense thus requires not only the ability to coordinate, but also a shared reading of what the first attack reveals.

The theory paper is complete and available as a working paper, and we are preparing it for submission. We are also preparing a companion laboratory experiment that varies the cost of aggression and the prior probability of a broad motive; I developed its treatment design, hypotheses, and planned analysis. Belief elicitation and conditional decisions will help distinguish changes in perceived danger from changes in willingness to resist. Planned extensions are described under Future Research.

Team Incentives, Observability, and Pay Transparency

Organizations can reduce the theoretical cost of teamwork by changing what workers observe, but those savings materialize only if workers make the inferences and choices on which the mechanism relies. In *Incentivizing Teamwork: An Experiment*, a project I initiated and lead in collaboration with Timothy Cason, we examine this tension in a setting where two workers undertake costly, complementary tasks and compensation can depend only on aggregate output.

The principal offers outcome-contingent bonuses and chooses an information structure along two dimensions: whether effort decisions are simultaneous or sequential, and whether bonus contracts are public or privately mediated. These dimensions capture two distinct forms of organizational transparency: the observability of coworkers' effort and the informativeness of their compensation. Four mechanisms uniquely implement joint effort under standard preferences. A fifth, symmetric-bonus benchmark supports joint effort but also admits mutual shirking, leaving the coordination problem unresolved.

The four mechanisms rely on different forms of strategic reasoning. Simultaneous public contracting uses asymmetric bonuses: one worker has a dominant incentive to work, allowing the other to infer that working is optimal. Simultaneous mediation uses privately observed bonuses and uncertainty about relative bonus ranks to reduce expected compensation while restoring ex ante pay symmetry. Sequential public contracting allows the later worker to condition effort on the earlier worker's action, creating endogenous peer monitoring. Sequential mediation jointly randomizes move order and bonuses, conceals the realized branch at the decision stage, and restricts continuation play after a first mover shirks; this eliminates the late-mover premium while preserving ex ante symmetry. Theoretical compensation costs across the five schemes satisfy a strict ranking, from highest to lowest: simultaneous public, simultaneous mediated, symmetric benchmark, sequential public, and sequential mediated.

Our pre-registered experiment asks whether these mechanisms achieve joint effort and deliver the predicted savings. We distinguish the frequency of joint effort, average realized principal expenditure, and cost-effectiveness, measured as joint-effort frequency divided by average expenditure. Under standard theory, all four uniquely implementing mechanisms yield joint effort, so lower costs translate directly into greater cost-effectiveness. In practice, mediation may increase reasoning demands, while sequential play may

facilitate peer monitoring but also activate reciprocity or spite. Incentivized belief reports help determine whether failures reflect pessimism about a teammate's effort or choices that depart from expected-money maximization given the worker's own reported belief. A simulation-based power analysis informed the pre-registration, and formal data collection at Purdue University is scheduled for spring 2027.

A complementary theoretical analysis introduces Fehr–Schmidt inequity aversion. It shows that the sequential public mechanism is relatively robust to concerns about unequal compensation, whereas mediation restores ex ante symmetry while still producing unequal realized payoffs.

The project evaluates incentive systems by what they achieve relative to what they actually cost. Greater observability and greater opacity can each reduce theoretical costs, but they place very different demands on the people using the institution: one asks workers to respond to what they see, the other to reason about what they cannot see.

Social Counterfactuals and Correlation Sensitivity

Dissatisfaction with an outcome may reflect a missed personal opportunity, a missed opportunity for a more favorable social comparison, or both; distinguishing these motives matters for understanding how people respond to information about alternatives. In *Correlation Sensitivity in Social Context: Theory and Experiment*, coauthored with Hendrik Rommeswinkel, we develop a framework that separates these two counterfactual channels.

The framework distinguishes regret over a forgone personal alternative from social regret over a forgone difference between one's own and another person's outcome. Social regret is not simply dissatisfaction with realized inequality: it concerns how the realized social comparison differs from one that could have occurred. This distinction allows us to represent behavior associated with fear of missing out in different layers of counterfactual comparisons.

We are developing an experimental task that varies the correlation between one's own and another person's outcomes while holding the marginal distributions over allocations constant, allowing the two channels to be identified separately. Hendrik leads the axiomatic analysis, and I lead the experimental design and implementation; the design and pre-analysis plan have been drafted, and the experiment is in preparation. The project complements my work on institutions by examining another reason that material outcomes alone may not explain behavior: people may evaluate those outcomes against different personal and social alternatives.

Future Research

My immediate priorities are to complete the revision of my job market paper and the data collection for the divide-and-conquer, teamwork, and social-counterfactual experiments. Beyond these, I will study how institutional and informational interventions affect strategic behavior: through which channels they achieve their intended effects, and when participants' responses redirect or offset those effects. A lesson common to my current projects is that a well-designed incentive can still fail, either because it changes another participant's incentives in unintended ways or because it requires inferences that participants do not draw. I will identify which design features create these demands and which modifications relieve them.

In my job market paper, the safety trap arises because the same parameter punishes the sender and raises the receiver's safe payoff. A natural next step is to separate the two, testing whether stronger discipline produces more informative communication once it no longer makes the alternative to trusting more attractive. A related design lowers the sender's payoff from the preferred action when interests conflict, asking whether truth-telling can become Pareto-superior to blind trust. Belief elicitation and separate measures of honesty and trust will distinguish changes in sender behavior from changes in receivers' willingness to act on messages. The broader question is when relaxing one participant's incentive constraint improves cooperation, and when the outcome remains limited by another participant's response.

Further phases of the divide-and-conquer experiment will study interventions that change the informational meaning of the first attack. One phase lets the attacker promise safety to the bystander and varies the

cost of breaking that promise. Another introduces records of past behavior, such as prior provocation or disclosed choices from a trust game played before the conflict.

In teamwork, low effort under a theoretically effective contract could reflect difficulty reasoning through a teammate's choices. I plan to vary reasoning support while preserving the relevant material incentives, testing whether simplifying strategic reasoning restores a mechanism's effectiveness.

Across these projects, theory will identify the incentives and inferences an institution requires, and experiments will test whether participants act on them and which design changes help. By identifying the conditions on which an intervention's success depends, I aim to contribute to behavioral game theory, experimental economics, and institutional design with explanations that guide both the evaluation of interventions and their adaptation to new strategic environments.